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DOOR HANDLE

Abstract

A door handle may have at least a handle element, a deflection element, a transmission element, a locking mechanism for locking a lock, and a handle support, to which the handle element and the deflection element are articulated. The transmission element may be coupled to the deflection element, so that upon actuation of the handle element, the deflection element can be actuated, and the lock can be unlocked thereby via the transmission element. The locking mechanism may have at least one mass lock, which, due to an accident, can be displaced into a locking position on the deflection element due to mass inertia. The locking mechanism may have a blocking device which irreversibly blocks the mass lock in the locking position.

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Background/Summary

FIELD

[0001] The invention relates to a door handle for a vehicle.

BACKGROUND

[0002] In order to prevent a door lock of the door handle from being unintentionally unlocked, for example in the event of an accident when the door is subjected to a sudden impact from the outside, it is known to provide a locking mechanism in the form of a mass lock which prevents the door lock from being accidentally opened or unlocked.

[0003] The invention is based on the object of providing a door handle for a vehicle with an improved locking mechanism, which prevents unwanted unlocking of a door lock.

[0004] The object is achieved according to the invention by a door handle having the features of claim 1.

SUMMARY

[0005] The door handle according to the invention comprises at least a handle element, a deflection element, a transmission element, a locking mechanism for locking a lock, and a handle support, to which the handle element and the deflection element are articulated, wherein the transmission element is coupled to the deflection element, and wherein, upon actuation of the handle element by means of the handle element, the deflection element can be actuated and the lock can be unlocked thereby via the transmission element, wherein the locking mechanism comprises at least one mass lock, which, due to an accident, in particular in the event of an accident or an impact, can be displaced into a locking position on the deflection element due to mass inertia, and wherein the locking mechanism further comprises a blocking device, which irreversibly blocks the mass lock in the locking position.

[0006] Irreversible blocking in the locking position means that the mass lock does not pivot back on its own, but rather remains in the blocked position so that, for example, the door handle and thus a vehicle door do not open in the event of an accident. After the accident, the mass lock can be reset, for example in a garage. This means that the mass lock, for example a blocking spring such as a leaf spring, can be returned to its starting position (rest position).

[0007] The advantages achieved by the invention are in particular that a mass lock that has been triggered and is in its locking position can be blocked so that it does not become disengaged or move out of its locking position during an accident. In other words, the mass lock remains locked in its locking position by means of the blocking device. This can ensure that the vehicle door is prevented from being opened from the outside during an accident, but also after an accident.

[0008] In the normal position or rest position (starting position), the mass lock is positioned relative to the deflection element such that the mass lock does not block the deflection element and thus the transmission element coupled thereto. This makes it possible to lock and unlock the lock, in particular a door lock, by actuating the handle element.

[0009] If the door or door handle is subjected to mechanical stress, for example in the event of an accident such as a side impact, a bearing frame of the door handle and/or the mass lock may be pressed in or acted upon by mass forces or acceleration forces, whereby the mass lock moves dynamically due to its mass inertia, in particular pivots about an axis of rotation, as a result of which a locking surface of the mass lock comes into engagement with a locking opening of the deflection element and thus moves into the locking position on the deflection element. As a result, the mass lock blocks any further movement of the deflection element and thus of the transmission element in the unlocking direction so that unlocking of the lock and thus opening of the door handle is prevented.

[0010] As a result of the automatic movement of the mass lock, the blocking device, in particular a

spring-loaded leaf spring, moves into a blocking position so that the mass lock is blocked in its locking position, in particular irreversibly.

[0011] In a possible development, the mass lock may comprise a unidirectional locking element and a mass element. The locking element and the mass element can be mounted so as to be rotatable about a common locking axis. For example, the locking element and the mass element can be mounted on a handle support or on a bearing bracket so as to be rotatable about the common locking axis.

[0012] Preferably, the locking element and the mass element are designed as a one-piece mass locking element. Alternatively, the locking element and the mass element may be designed separately and firmly connected to one another. In particular, the mass lock has only one locking position.

[0013] In addition, the deflection element can have a locking opening, into which the mass lock engages in a locking manner in the locking position. When the mass lock is brought into its locking position on the deflection element, the blocking device can, preferably automatically, move into a blocking position on the mass lock in order to irreversibly block the mass lock in its locking position.

[0014] The blocking device can be designed as a leaf spring, for example. The leaf spring can, for example, comprise a spring leg, which, in particular in a rest position (also called a starting position or an idle position) of the mass lock, rests with an abutment portion laterally on the mass lock in a spring-loaded manner.

[0015] When the mass lock is brought into its locking position, for example in the event of an accident, when accelerations, in particular acceleration forces, act on the vehicle and a vehicle door with the door handle, for example in the event of a side impact, the spring leg can move automatically from its lateral abutment on the mass lock into the blocking position on the mass lock. The spring leg expands and automatically moves into the blocking position on the mass lock. For example, the spring leg can comprise an extension, which, in the blocking position, irreversibly blocks the mass lock.

[0016] The deflection element can be designed as a disk-shaped locking body, in which the locking opening is arranged, into which the locking element pivots or engages. The deflection axis and the locking axis are preferably arranged perpendicular to one another.

Description

DESCRIPTION OF THE FIGURES

[0017] Exemplary embodiments of the invention are explained in greater detail with reference to the drawings, in which:

[0018] FIG. 1 schematically shows a door handle having a mass lock and a blocking device,

[0019] FIG. 2 schematically shows a perspective view of the mass lock and the blocking device in their rest positions or starting positions,

[0020] FIG. 3 schematically shows a further perspective view of the mass lock and the blocking device in their rest positions or starting positions,

[0021] FIG. 4 schematically shows a perspective view of the mass lock in its locking position and the blocking device in its blocking position, and

[0022] FIG. 5 schematically shows a further perspective view of the mass lock in its locking position and the blocking device in its blocking position.

DETAILED DESCRIPTION

[0023] Parts corresponding to one another are provided with the same reference signs in all the drawings.

[0024] FIG. 1 schematically shows a door handle 1 for a vehicle (not shown).

[0025] The door handle **1** is in particular an external door handle. The door handle **1** usually comprises a housing frame, which can in particular comprise a bearing frame or a handle support **2** for a handle element **4**. The handle support **2** can be fastened to a door panel or another bearing bracket in a manner not shown in detail.

[0026] The handle element **4** is, among other things, articulated to the handle support **2**. The handle element **4** is designed, for example, as a pull handle, a rotary handle, or a folding handle.

[0027] In addition, a deflection element **6** can be articulated to the handle support **2** and/or to the bearing bracket. The deflection element **6** is designed, for example, as a deflection lever, a deflection bracket, a deflection pulley, or the like. The deflection element **6** is mounted on the handle support **2** or on the bearing bracket so as to be pivotable about a deflection axis **8**.

[0028] The handle element **4** and the deflection element **6** are in particular coupled to one another such that, upon actuation of the handle element **4**, the deflection element **6** is also actuated. For example, the handle element **4** and the deflection element **6** can be designed in one piece. In particular, they can be designed as a molded component. Alternatively, the handle element **4** and the deflection element **6** can be designed as separate components.

[0029] A transmission element **10** is coupled to the deflection element **6**. The transmission element **10** is designed, for example, as a Bowden cable.

[0030] A lock **12** is coupled to the end of the transmission element **10** opposite the deflection element **6**, which lock can be actuated, in particular unlocked, via the deflection element **6** and the transmission element **10** upon actuation of the handle element **4**.

[0031] Furthermore, a locking mechanism **14** is provided, which is designed such that it has a locking position **100** (shown in FIGS. **4** and **5**), in which the deflection element **6** is or will be locked in a rest position **300** (shown in FIGS. **2** and **3**) or a starting position, and a blocking position **200** (shown in FIGS. **4** and **5**), in which the locking position **100** is or will be held securely.

[0032] In the locking position **100**, the locking mechanism **14** thus blocks pivoting of the deflection element **6** into an unlocking position for the lock **12**. By securely holding the locking position **100** in the blocking position **200** of the locking mechanism **14**, the deflection element **6** remains blocked and the lock **12** is thus securely locked.

[0033] The locking mechanism **14** comprises at least one mass lock **14.1**, which, in the event of an accident or impact, can be displaced into the locking position **100** on the deflection element **6** due to mass inertia, in particular automatically, for example by means of a stored force, for example a spring force, and wherein the locking mechanism **14** further comprises a blocking device **14.2**, which irreversibly blocks the mass lock **14.1** in the locking position **100**.

[0034] Such a locking mechanism **14** makes it possible in a simple manner for a mass lock **14.1** that has been triggered and is in its locking position **100** to be or remain blocked so that it cannot become disengaged or move out of its locking position **100**. This can ensure that the vehicle door is prevented from being opened from the outside during an accident, but also after an accident.

[0035] FIGS. **2** and **3** schematically show a perspective view of the mass lock **14.1** and the blocking device **14.2** in their rest positions **300** (called starting positions or normal positions).

[0036] In the rest position **300** of the mass lock **14.1**, it is positioned relative to the deflection element **6**, in particular at a distance therefrom, such that the mass lock **14.1** does not block the deflection element **6** and thus the transmission element **10** coupled thereto (shown in FIG. **1**). It is thus possible to lock and unlock the lock **12** (shown in FIG. **1**), in particular a door lock, by actuating the handle element **4** (shown in FIG. **1**).

[0037] The deflection element **6** is designed, for example, as a deflection pulley, in particular a double pulley, with a guide seat **6.1** for the transmission element **10**. The deflection element **6** is mounted on the handle support **2** or bearing bracket so as to be pivotable about the deflection axis **8**. Arrow **400** indicates the unlocking direction of the unlocking movement of the deflection element **6** upon actuation of the handle element **4** in order to actuate, in particular to unlock, the

lock **12** via the transmission element **10**.

[0038] The deflection element **6** has a locking opening **6.2**, in particular a through-opening, into which the mass lock **14.1** can engage in the locking position **100**. In FIGS. **2** and **3**, the mass lock **14.1** has not engaged in the locking opening **6.2**.

[0039] The mass lock **14.1** is spring-loaded, for example by means of a spring element **14.1.3**, on the handle support **2** (shown in FIG. **1**) and/or on the bearing bracket or bearing frame. The mass lock **14.1** comprises a mass element **14.1.1** and a locking element **14.1.2**. The mass lock **14.1** can be designed in one piece. The mass element **14.1.1** and the locking element **14.1.2** can be designed as a molded component. Alternatively, the mass element **14.1.1** and the locking element **14.1.2** may be separate components that are firmly connected to one another.

[0040] The mass lock **14.1** together with the locking element **14.1.2** is pivotally mounted on the handle support **2** or bearing frame or bearing bracket by means of a pivot bearing **14.3**. In other words, the mass lock **14.1** together with the locking element **14.1.2** can engage in the locking position **100** (shown in FIGS. **4** and **5**) on the deflection element **6**.

[0041] The blocking device **14.2** is designed, for example, as a leaf spring **14.2.1** (also called flat spring). The leaf spring **14.2.1** can, for example, have a U-shape or an omega shape, or the like.

[0042] In the rest position **300** shown, the blocking device **14.2** is not in blocking engagement with the mass lock **14.1**. In particular, the leaf spring **14.2.1** can be arranged in portions laterally of the mass element **14.1.1**. At least one portion of a spring leg **14.2.2**, for example, rests laterally on the mass lock **14.1** in a spring-loaded manner. In particular, the spring leg **14.2.2** extends perpendicularly to the locking axis **14.3.1** of the pivot bearing **14.3** of the mass lock **14.1** and at least in portions with an abutment portion **14.2.4** along an outer side of the mass lock **14.1**, in particular of the mass element **14.1.1**.

[0043] For holding the blocking device **14.2** on the handle support **2** or the bearing frame, the leaf spring **14.2.1** can have holding elements **14.2.3**, in particular holding lugs or the like.

[0044] FIGS. **4** and **5** each schematically show a perspective view of the locking mechanism **14** with the mass lock **14.1** in its locking position **100** on the deflection element **6** and the blocking device **14.2** in its blocking position **200** on the mass lock **14.1**.

[0045] If the door or the door handle **1** (shown in FIG. **1**) is subjected to mechanical stress, for example in the event of an accident such as a side impact, the handle support **2** of the door handle **1** and/or the mass lock **14.1** may be pressed in or acted upon by mass forces or acceleration forces, whereby the mass lock **14.1** moves dynamically due to its mass inertia, in particular pivots about the locking axis **14.3.1** (=axis of rotation).

[0046] As a result, a locking surface, in particular the locking element **14.1.2** designed as a hook or nose, of the mass lock **14.1** comes into engagement with the locking opening **6.2** of the deflection element **6** and thus moves into the locking position **100** on the deflection element **6**. In particular, the mass lock **14.1** pivots into the locking opening **6.2** in the engagement direction **500**. As a result, the mass lock **14.1** blocks movement of the deflection element **6** and thus of the transmission element **10** in the unlocking direction according to arrow **400** so that unlocking of the lock **12** and thus opening of the door handle **1** is prevented.

[0047] As a result of the movement of the mass lock **14.1**, the blocking device **14.2**, in particular the spring-loaded leaf spring **14.2.1**, moves into its blocking position **200** so that the mass lock **14.1** is irreversibly blocked in its locking position **100**.

[0048] During the pivoting movement of the mass lock **14.1**, the spring leg **14.2.2** moves from its lateral abutment on the mass element **14.1.1** so that the leaf spring **14.2.1** can expand.

[0049] An extension **14.2.5** protruding from the abutment portion **14.2.4** moves underneath the mass lock **14.1** and blocks it perpendicularly to the engagement direction **500**. The abutment portion **14.2.4** blocks the mass lock **14.1** against the engagement direction **500** (also called locking direction). In other words, the abutment portion **14.2.4** ensures that the mass lock **14.1** cannot be moved from its locking position **100**. The mass lock **14.1** is thus irreversibly blocked in its locking

position **100** by means of the engaged blocking device **14.2**, in particular the leaf spring **14.2.1**, and cannot be released.

LIST OF REFERENCE SIGNS

[0050] **1** Door handle [0051] **2** Handle support [0052] **4** Handle element [0053] **6** Deflection element [0054] **6.1** Guide seat [0055] **6.2** Locking opening [0056] **8** Deflection axis [0057] **10** Transmission element [0058] **12** Lock [0059] **14** Locking mechanism [0060] **14.1** Mass lock [0061] **14.1.1** Mass element [0062] **14.1.2** Locking element [0063] **14.1.3** Spring element [0064] **14.2** Blocking device [0065] **14.2.1** Leaf spring [0066] **14.2.2** Spring leg [0067] **14.2.3** Holding element [0068] **14.2.4** Abutment portion [0069] **14.2.5** Extension [0070] **14.3** Pivot bearing [0071] **14.3.1** Locking axis [0072] **100** Locking position [0073] **200** Blocking position [0074] **300** Rest position [0075] **400** Arrow [0076] **500** Engagement direction

Claims

1. A door handle, comprising: a handle element, a deflection element, a transmission element, a locking mechanism for locking a lock, and a handle support, to which the handle element and the deflection element are articulated, wherein the transmission element is coupled to the deflection element, and wherein, upon actuation of the handle element, the deflection element can be actuated and the lock can be unlocked thereby via the transmission element, wherein the locking mechanism comprises at least one mass lock, which, due to an accident, can be displaced into a locking position on the deflection element due to mass inertia, and wherein the locking mechanism further comprises a blocking device, which irreversibly blocks the mass lock in the locking position.
 2. The door handle according to claim 1, wherein the mass lock comprises a unidirectional locking element and a mass element.
 3. The door handle according to claim 2, wherein the locking element and the mass element are mounted so as to be rotatable about a common locking axis.
 4. The door handle according to claim 2, wherein the locking element and the mass element are designed as a one-piece mass locking element.
 5. The door handle according to claim 1, wherein the deflection element has a locking opening, into which the mass lock engages in a locking manner in the locking position.
 6. The door handle according to claim 1, wherein, when the mass lock is brought into its locking position on the deflection element, the blocking device moves into a blocking position on the mass lock in order to irreversibly block the mass lock in its locking position.
 7. The door handle according to claim 1, wherein the blocking device is designed as a leaf spring.
 8. The door handle according to claim 7, wherein the leaf spring comprises a spring leg, which rests with an abutment portion laterally on the mass lock in a spring-loaded manner.
 9. The door handle according to claim 8, wherein, when the mass lock is brought into its locking position, the spring leg automatically moves from its lateral abutment on the mass lock into the blocking position on the mass lock.
 10. The door handle according to claim 9, wherein the spring leg comprises an extension, which, in the blocking position, irreversibly blocks the mass lock in its locking position.
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